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A novel approach for the fabrication of planar waveguides from heavy metal oxide glasses

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ABSTRACT

An extrusion process is used to fabricate preforms for the development of planar waveguides in glass ribbon format through a stretching process. The feasibility of the overall process is demonstrated using three tellurite glass compositions. The manufactured preforms present multilayered structures with the higher refractive glass layer thickness of the order of 40 μm. The longitudinal and transversal variations of the layer thicknesses are discussed with respect to waveguiding application tolerances. Advantages, limitations and improvement of the technique are also discussed. 3 mm wide multilayered planar waveguides in ribbon geometry are produced by stretching the preforms by a 1:5 ratio over a length of 400 mm. The waveguides were examined by optical and electron microscopy and no signs of crystallization or contamination are observed. Optical waveguiding is successfully assessed in 7.4 μm thick and 9 mm long film waveguides.

1. Introduction

Over the last two decades, numerous reports have been published on the exciting prospect of using mid- and far-infrared transmitting glasses for compact integrated optic devices. The possible range of applications for such devices is broad, going from optical amplifiers, lasers and all-optical switching to thermal, chemical and biological sensors.

The typical first fabrication step of a planar step-index waveguide, or more complex optical circuitry, involves the manufacture of one or more thin films of material on a suitable substrate. The basic requirement is that the thin film material forming the guiding layer possesses a higher refractive index (RI) than that of the cladding materials in order to ensure light confinement within the film. Various film fabrication techniques have been applied to fabricate such structures from Heavy Metal Fluoride glasses (HMFG), Heavy Metal Oxide glasses (HMOG) and chalcogenide glasses. Some common deposition techniques arise from technologies such as sputtering [1], thermal vapor deposition [2–4] or Pulsed Laser Deposition [5,6]. Some other techniques are more particular to glass science technologies such as ion exchange [7,8] and spinning of sol–gel [9]. However, because of the multicomponent nature of the above glass systems, each of the above approaches brings a range of limitations such as film stoichiometry, homogeneity, film mechanical integrity and range of usable glass compositions. Also, processing time and equipment cost remain other important issues towards potential commercial development.

The possibility to manufacture planar waveguides in ribbon geometry by stretching a planar preform as an analogous process to optical fiber drawing could overcome the above limitations. The concept is not new, as early as 1973 Laybourn reported the fabrication of a ribbon waveguide from a single glass composition [10]. In 1992, Corning Inc. patented the concept of multilayered core/clad structured ribbon from silicate glasses [11]. Using a different preform fabrication technique, a similar silica glass layered ribbon structure was recently reported by Webb et al. under the name of flat fiber [12].

Producing film waveguide in ribbon geometry by means of stretching a preform implies that all the fabrication constraints are defined by the preform manufacturing process. To be viable the latter process would need to meet the following requirements:

- versatile in terms of glass composition;
- retain glass stoichiometry and glass physical properties;
- produce a layered structure with guiding layer(s) in vicinity of top surface;

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- enable production of layer(s) of the order 10 s of μm thick and
- reasonable processing time.

The extrusion process has been used to produce core/clad structure optical fiber preform from HMFG and chalcogenide glasses. In this process, two bulk glass samples of cylindrical shape and of different compositions (and hence refractive indices) are placed into the chamber of an extruder having a circular output die. The glass having the higher RI is placed on top. Typically, the whole assembly is heated up to a temperature corresponding to a glass viscosity of about 10^8 Pa s. A pressure of several 10 s of MPa is then applied onto the glass billets. The flow velocity gradient generated between the extrusion die wall and the center of the die results in material from the central region of the high RI glass billet exiting the die before the low refractive index glass which is in vicinity of the die wall. The extruded rod then consists of a high refractive index glass material surrounded by a layer of low refractive index material. That is to say, a core/cad structured preform for optical fiber.

Some of the optical fibers produced from such a preform have been reported to have losses as low as 0.01 dB/m [13]. This latter figure suggests that the extrusion technique is able to produce preforms with a suitable core/cladding interface quality for short length devices whilst also maintaining a low contamination level. The main aim throughout the work presented in this paper was to investigate an extrusion route to fabricate preforms for making waveguiding thin films in ribbon format.

We present our first achievements using tellurite glasses for making planar waveguides by stretching the fabricated preforms. At this early development stage, we comment on the potential of such an approach and its eventual limitations.

2. Experimental

2.1. Glass synthesis and characterization

Three glass compositions (see Table 1) were synthesized to carry out this investigation, namely TZN2, TZN3 as high RI glasses for light confinement and TZN1 as the low RI cladding substrate. Chemical precursors (ALFA AESAR, PURATRONIC) were weighed into an Au crucible (Birmingham Metal Ltd.) which was placed inside a vertical-bore resistance furnace (Instron, SFL). The melting process was carried out under a dried O_2 atmosphere at 800 °C for 3 h and the temperature was then decreased to 750 °C and held isothermally for 1 h prior to casting. The melts were cast into brass moulds, preheated to $[T_g - 10]$ °C (T_g is glass transformation temperature). Cast glasses were then annealed at $[T_g - 10]$ °C for 3 h and cooled down to room temperature at 1 °C /min. T_g and onset crystallization temperature, $T_{x,}$ of TZN1, TZN2 and TZN3 were measured using a commercial Differential Thermal Analyser (DTA, Perkin Elmer, Pyris DTA7).

The viscosity (η)/temperature curves of TZN1, TZN2 and TZN3 were measured using the parallel plate technique [14]. Measurements were performed using a commercial Differential Mechanical Analyser (DMA, Perkin–Elmer Pyris DMA 7). The measurement technique was calibrated using the standard viscosity reference glass material 717A supplied by the National Bureau of Standards, USA. The corresponding error of the measurement was ±5 °C over the range of viscosity 10^6 Pa s to $10^{8.5}$ Pa s. Measurements were performed by applying a load of 100 mN at a heating rate of 5 °C/min. The compression of the sample was recorded with respect to time and the viscosity of the supercooled melt was calculated following the procedure described in Ref. [14] using the Vogel–Fulcher–Tammann (VFT) equation [15,16]. The refractive indices of TZN1, TZN2 and TZN3 were measured at a wavelength of 632.8 nm using a commercial prism coupler (Metricon, model 2010).

2.2. Preform fabrication

In-house constructed equipment was used to extrude bulk glass into the preform shape. To demonstrate the versatility of the fabrication process, two types of preforms were manufactured, referred to as Preform 1 and Preform 2. In both cases the extrusion die aperture was a rectangular slit 2.5 mm × 17 mm. The die material was stainless steel.

For Preform 1, two cuboid glass bulk samples, of compositions TZN1 and TZN3, respectively, were loaded into the extrusion chamber. The extrusion was carried out downwards. The higher refractive index glass (TZN3) was placed at the bottom of the extrusion chamber with the lowest RI glass (TZN1) on top, the configuration is shown in Fig. 1(a). This is a reverse configuration to the one used for optical fiber preform fabrication where the lowest RI glass would be placed at the bottom of the extrusion chamber. One common feature of optical fiber preforms produced by extrusion is the reported high ratio between the core glass (higher RI value) diameter ($D1$) and the overall diameter ($D2$) of the preform. Typically $D1/D2 = 0.85$ has been observed, and such a ratio has been maintained within ±0.05 over 200 mm length [17]. From this value, one can therefore anticipate for Preform 1 a cross-section profile such as the schematic shown in Fig. 1(b), with a high value for the ratio $H1/H2$. Please note the xyz axis orientation in both Figs. 1(a) and (b).

Multilayer planar waveguides offer the framework upon which to design a number of optical devices such as couplers and

Table 1
Batch compositions of the tellurite glasses melted in bulk and then used to fabricate Preforms 1 and 2, which were stretched to manufacture planar waveguide in ribbon format.

	Mol%			Wt%			
	TeO_2	ZnO	NaO_2	Er_2O_3	Ce_2O_3	Tm_2O_3	Yb_2O_3
TZN1	75	15	10	–	–	–	–
TZN2	80	10	10	–	–	1	1.5
TZN3	80	10	10	1	1.5	–	–

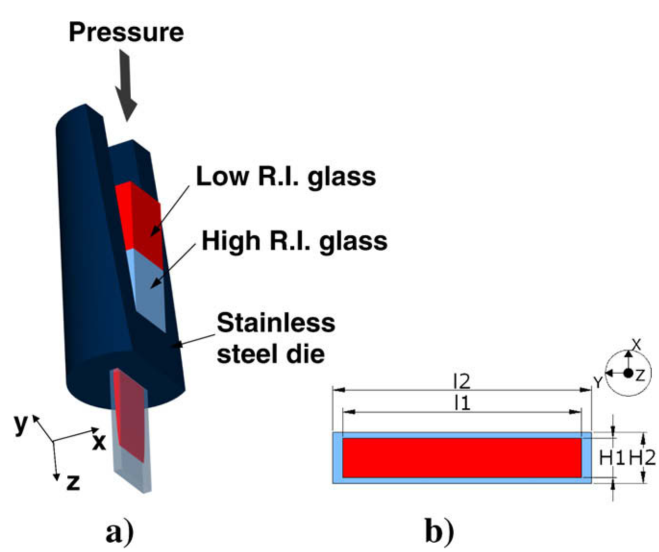

Fig. 1. (a) Schematic diagram of the extrusion set-up used for the fabrication of Preform 1. Please note that for Preform 2 an additional high refractive index glass was used, the rest of the set-up was identical. (b) Schematic of the cross-section profile expected for Preform 1.

switches. A low refractive index layer on top of the optical guiding layer is also beneficial both for stable operation of the waveguide and for packaging purposes. To investigate the possibility of fabricating multilayered preforms via extrusion, an additional cuboid bulk glass of composition TZN2 was placed in the extrusion chamber between the two glass blocks of compositions TZN1 and TZN3 for the fabrication of Preform 2.

A waveguide structure involving layers of TZN2 and TZN3 glasses could be exploited for broadband emission and lasing applications in the 2 μm region. However at this stage, the use of the TZN2 glass composition in the extrusion process was first and foremost motivated by the compatibility of this glass with the TZN1 and TZN3 glass compositions in terms of its thermo-mechanical properties. Moreover, the presence of Yb^{3+} and Tm^{3+} rare earth ions in glass composition TZN2 eases the identification of the individual layers within the preform and the waveguide through microscopy and optical characterization.

The extrusion process was carried out at 320 °C, which corresponds to a viscosity of 10^8 Pa s based on data resulting from viscosity measurements detailed in Section 3.1. Preforms 1 and 2 were both extruded at a speed of 1.5 mm/min.

2.3. Preform stretching for planar waveguide fabrication

The stretching of the preforms was carried out on an in-house designed fiber drawing tower. A resistance furnace was used whose inner diameter was 18 mm. In order to avoid any contact of the preform with the furnace wall during the initial neck-down process, the width of each of Preform 1 and Preform 2 was reduced to 15.4 mm and 15.2 mm, respectively, by polishing 1 mm away along each lateral side of the preforms. The stretching process was carried out under the same conditions for both Preforms 1 and 2. The preform was fed downwards into the furnace at a speed of 1 mm/min and stretched at a speed of 25 mm/min. These latter figures corresponded to a stretching ratio of 1:5 in order to obtain ribbons of the order of 3 mm wide from Preforms 1 and 2. Ribbon 1 and Ribbon 2 resulted from the stretching of Preform 1 and Preform 2, respectively.

2.4. Preforms and ribbons characterization

Short sections (≈3 mm long) of Preforms 1 and 2 were cut and polished to examine the preform cross-sections and to establish the thicknesses of the high refractive index layers by optical microscopy. Tellurite glasses are amenable to optical microscopy being transparent across the visible range of the spectrum. The quality of the interface between the layers of different glass composition was also investigated for Preform 2 by scanning electron microscopy (XL30 Philips ESEM) in backscattering mode.

12 mm long sections of Ribbons 1 and 2 were cut and polished along their cross-section to a 1 μm finish using METADII® diamond paste and polishing lubricant (Buehler Coventry, UK). The length of the samples after polishing was 9 mm. Optical microscopy and SEM techniques were used to measure the thickness of the waveguiding layers and the quality of the interface between the glass layers.

2.5. Optical waveguide characterization

Optical waveguiding of 9 mm long, ribbon waveguide samples was assessed by the end-fire technique at 632.8 nm wavelength (He–Ne laser, Uniphase model 1108p, 0.95 mW). The laser beam was focused on the input cross-section of the film waveguide using a ×25 microscope objective (Melles Griot, numerical aperture (NA) of 0.5). After propagation down the waveguide, the light was collected using a ×10 microscope objective (Melles Griot, NA of 0.6). The light was then focused onto a Siemens XQ1112 infrared vidicon tube installed in a 109A CCTV camera, from which the output signal was acquired on a computer to image the near-field profile distribution at the waveguide output face.

3. Results and discussion

3.1. Glass synthesis and characterization

Characteristic temperatures T_g, T_x and the Hruby parameter [18] of glass stability on reheating above T_g: $\Delta T = (T_x - T_g)$ are shown in Table 2 for TZN1, TZN2 and TZN3. The values of T_g were similar for all three glass compositions, 286 ± 2 °C, 284 ± 2 °C and 285 ± 2 °C, respectively. The crystallization temperature was found to be lower for TZN2 and TZN3 (416 ± 5 °C and 410 ± 5 °C respectively) than for TZN1 (430 ± 5 °C), which contained a higher ZnO content and lower TeO_2 content. A maximum Hruby glass stability value of 144 ± 7 °C was found for TZN1 whilst the minimum value was 123 ± 7 °C for TZN3.

The viscosity curves of TZN1, TZN2 and TZN3 glasses as a function of temperature are shown in Fig. 2. The temperatures corresponding to a viscosity of 10^{12} Pa s *i.e.*, approximately to the glass transformation region were found to be 278 ± 5 °C, 282 ± 5 °C and 284 ± 5 °C for TZN1, TZN2 and TZN3, respectively. These values were in good agreement with the DTA results taking into account experimental error on measurements indicating that DTA collecting under these conditions yields a value for T_g close to 10^{12} Pa s. Over the range of viscosity 10^5–10^9 Pa s, the viscosity values remained of the same order of magnitude at successive temperatures for the three glass compositions, for instance: $\eta = 10^{7.5}$ Pa s, $10^{7.8}$ Pa s and 10^8 Pa s at 320 °C, for TZN3, TZN2, and TZN1, respectively. Furthermore, the temperatures corresponding to the fiber drawing viscosity range (≈10^5 Pa s) remained about 50 °C below the DTA crystallization temperatures making these three glasses suitable for the manufacture of ribbon via the stretching process.

The refractive index of TZN2 and TZN3, at 632.8 nm, were found to be higher (2.046 ± 0.001 and 2.047 ± 0.001 respectively) than that of TZN1 (2.022 ± 0.001). This was due to the higher molar content of the more polarisable TeO_2 and lower molar content of less polarisable ZnO in TZN2 and TZN3 compared with the TZN1 composition; the molar concentration of Na_2O remained the same in all three glasses.

3.2. Preform fabrication

For the following sections, *xyz* coordinates refer to the two orthogonal cross-sectional axes (*x* is thickness and *y* depth of the multi-layer stack); *z* is length where *z* = 0 is first glass to exit the extruder.

3.2.1. Preform 1

A 197 mm long preform was extruded using TZN1 and TZN3 cuboid glass bulk samples. Because of swelling phenomena the width (17.4 ± 0.2 mm) and the thickness (2.9 ± 0.2 mm) of the preform

Table 2
Differential thermal analysis and refractive index (RI at 632.8 nm wavelength) measurements of the bulk-melted tellurite glasses whose compositions are presented in Table 1 *viz.:* extrapolated onset glass transition temperature T_g; onset crystallization temperature T_x.

	$T_g \pm 2$ (°C)	$T_x \pm 5$ (°C)	$\Delta T \pm 7$ (°C)	RI $_{\text{at 632.8 nm}} \pm 0.001$
TZN1	286	430	144	2.022
TZN2	284	416	132	2.046
TZN3	285	410	125	2.047

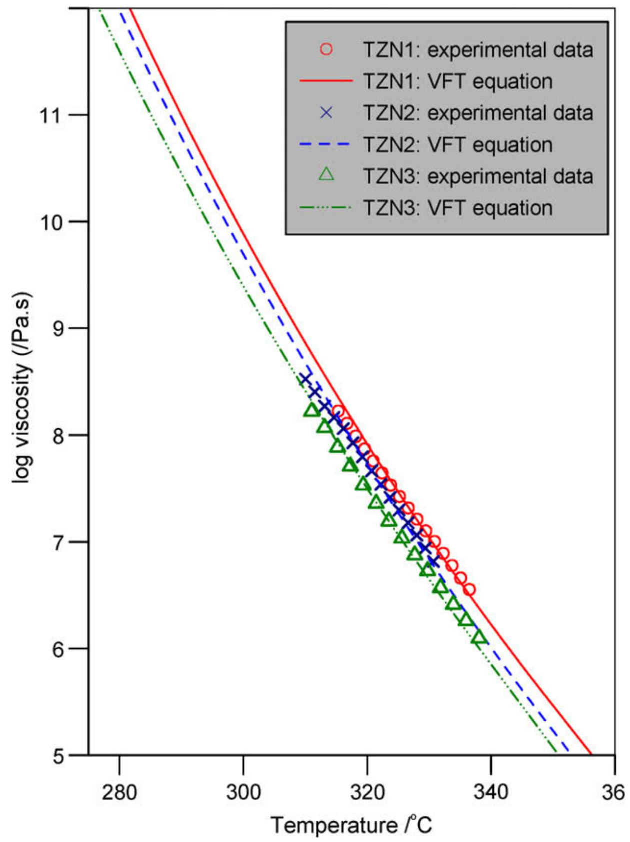

Fig. 2. Viscosity vs. temperature curves for glasses whose compositions are presented in Table 1. Experimental data obtained for 100 mN load and fitted curves against Vogel–Fulcher–Tamman (VFT) equation [15,16].

were larger than the corresponding die dimensions (17.2 ± 0.2 mm and 2.5 ± 0.2 mm). The preform was slightly bent in the (*yz*) plane, taken as a sign of a nonhomogeneous flow at the die output. Although the preform surface was mostly pristine, a pattern of surface striations could be observed along its length. Their origin was physical in nature and related to micro- defects at the output edge of the die. Non-homogenous flow and die defects are not inherent phenomena to the extrusion process but equipment related instead.

In Fig. 3, a composite picture shows part of a cross-sectional view of the preform at its end extremity that corresponds to the latest stage of the extrusion process ($z = 195$ mm). The overall shape and main features of the preform were almost symmetrical with respect to the ($Y'Y$) axis shown in Fig. 3.

One can see from Fig. 3 the TZN3 glass layer surrounding the TZN1 glass substrate, which constitutes most of the preform in this section. Also due to die-swelling phenomena, during the extrusion, and wall friction, the edges and corners of the preform presented a rounded, rather than a sharp, shape. The interface between the TZN1 and TZN3 glass appeared to be free of contamination under optical microscopy in the sections examined.

The minimum value of the TZN3 layer thickness layer was measured to be 34.8 ± 1 μm. The thickness variation of the TZN3 layer with respect to the position across the preform width is also shown in Fig. 3. The average value of the thickness was found to be 36.8 ± 1 μm in the central part of the preform with a maximum variation of ±2 μm (≈ ±6%) across about 8 mm. Outside this central area, and towards the edges of the preform, the thickness of the layer increased rapidly.

As shown in Fig. 4, for sections corresponding to $z = 91$ mm and $z = 143$ mm the minimum value of the TZN3 glass layer was measured to be 44.4 ± 1 μm and 166.3 ± 1 μm, respectively. The thickness variation with respect to the preform width for these sections is still to be analyzed.

In terms of longitudinal variation of the minimum thickness, the above figures translate to a decrease of the thickness by 70% between $z = 91$ mm and $z = 143$ mm and a decrease by 16% between $z = 143$ mm and $z = 196$ mm. Although these latter variation values might appear restrictive one must take into account that they were attenuated later through the stretching process, as now discussed.

Consider a linear variation of the thickness with respect to the preform length (z). A variation by 70% of the thickness of the TZN3 glass layer over 52 mm length of preform becomes 0.6%/cm for the same layer in a ribbon obtained by stretching this preform by a 1:5 ratio. For a 16% thickness variation over a 53 mm length of preform the variation of the guiding layer in the ribbon becomes

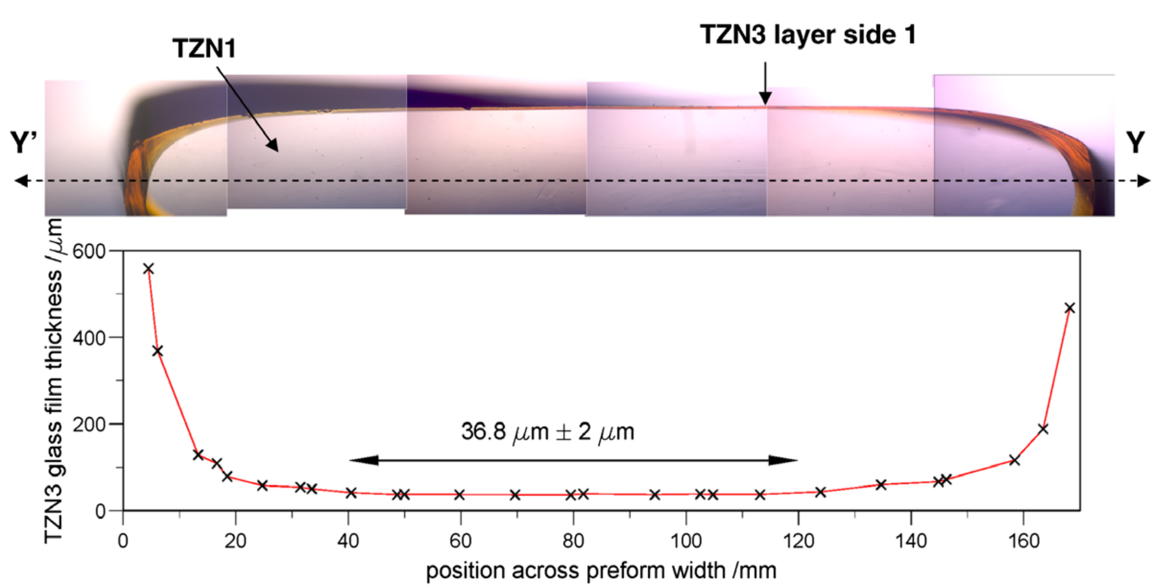

Fig. 3. Optical micrographs composite showing part of the cross-sectional profile of Preform 1 at the latest stage of the extrusion ($z = 196$ mm, in Fig. 4). The overall profile of Preform 1 is symmetric with respect to the $Y'Y$ axis. Below the corresponding TZN3 glass thickness is shown with respect to Preform 1 width (y).

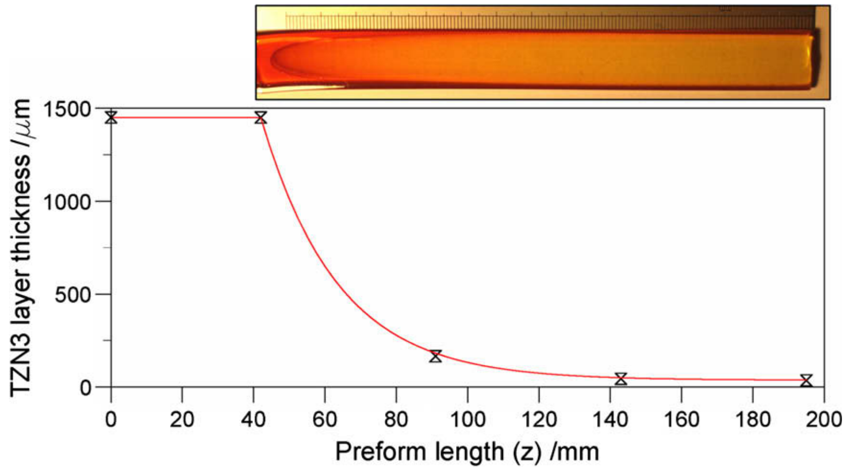

Fig. 4. Longitudinal variation of the minimum thickness of the TZN3 glass layer measured for several sections of Preform 1. The line is shown as a guide to the eye. A plan view photography of Preform 1 is also shown for reference.

0.15%/cm. It is anticipated that such tolerance values are acceptable in term of robust device design for commercial development of short planar devices.

3.2.2. Preform 2

TZN1, TZN2 and TZN3 cuboid glass bulks were extruded to manufacture Preform 2 which was 215 mm long. The same main features as those observed for Preform 1 were also observed for Preform 2. The overall cross-sectional profile of Preform 2 was similar to that of Preform 1. Swelling phenomena again occurred. The width of the preform was 17.2 ± 0.2 mm and the thickness was 2.86 ± 0.2 mm. Bending in the (yz) plane and the presence of surface striations were again observed as for Preform 1.

Fig. 5 presents a scanning electron micrograph captured in backscattering mode of the section of Preform 2 at $z = 157$ mm. No sign of contamination or heterogeneous crystallization was observed either at the interface between TZN3 and TZN2 or at the interface between TZN2 and TZN1 for this section of Preform 2. This suggests the possibility of manufacturing low loss planar waveguides by stretching an extruded preform, as demonstrated for optical fiber [13].

The minimum values of the thickness of the TZN2 and TZN3 glass layers in Preform 2 were measured at $z = 98$ mm, 157 mm, 215 mm. Preform 2 was cut at these three points to obtain two preforms about 50 mm long which is the maximum length of preform that could fit in the drawing tower furnace. These values have been plotted against z in Fig. 6. For both TZN3 and TZN2 glass layers, the general trend of thickness variation with respect to z was the same to that observed for the TZN 3 glass layer in Preform 1, see Section 3.2.1 and Fig. 4.

3.3. Ribbon thin film waveguide

3.3.1. Ribbon 1

The section of Preform 1 corresponding to $143 < z < 196$ mm was selected to be stretched into Ribbon 1. This particular section of Preform 1 was selected in order to achieve a ribbon with a minimum longitudinal thickness variation of the guiding layer as discussed in Section 3.2.1. Because of the initial necking down process, the ribbon width went through a transient regime where it increased from less than 1 mm towards the theoretical 3 mm permanent regime value. After the transient regime, the width of the ribbon could be maintained to 2.9 ± 0.4 mm over a length of about 400 mm. The stretching had to be stopped at this stage because of the setup configuration, which had been originally designed rather for optical fiber drawing. The discussion below focuses on the permanent regime section of the ribbon only. A top view photograph of this part of Ribbon 1 is shown in

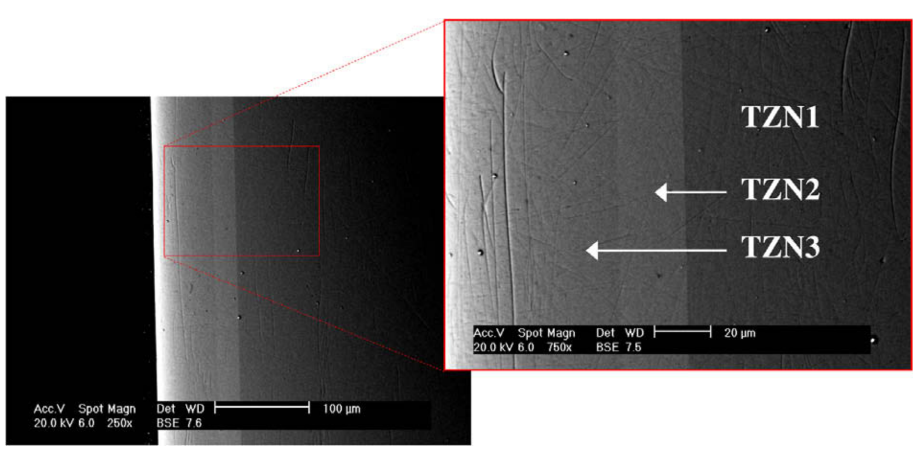

Fig. 5. Scanning electron micrographs captured in backscattering mode for the section of Preform 2 corresponding to $z = 157$ mm in Fig. 6.

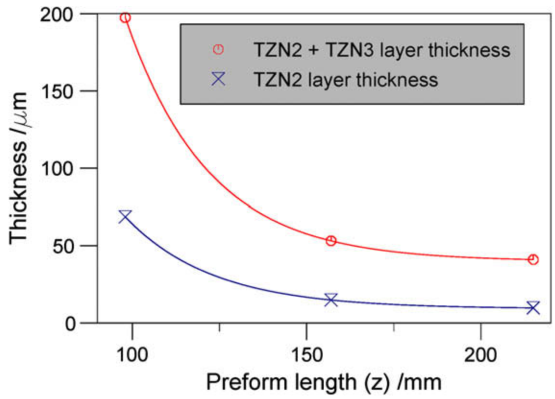

Fig. 6. Longitudinal variation of the minimum thickness of the TZN3 and TZN2 glass layers measured at three different sections of Preform 2. The line is shown as a guide to the eye.

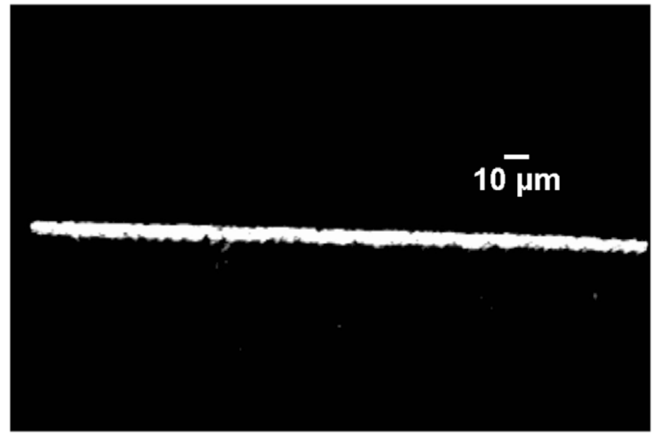

Fig. 8. Optical waveguiding assessment at 632.8 nm wavelength of a 9 mm long planar waveguide prepared from a mid-section of Ribbon 2. Please note that the input beam coupled into both the TZN2 and TZN3 glass layer. The apparently rather uneven edges of the waveguiding region are due to the preparation method of the endface of polishing.

Fig. 7(a). The width fluctuations which arise from stretching ratio fluctuations are believed to be equipment related. Stretching equipment optimized for ribbon making would certainly allow longer ribbons to be made by using the whole preform length and would provide a better control on the stretching ratio as well.

Fig. 7(b) shows an optical micrograph of the cross-section of Ribbon 1. Notice the conservation of the main features of Preform 1 throughout the stretching process *i.e.* a central area with a near uniform thickness and two lateral areas where the thickness increased rapidly towards the edges. It should be emphasized however, that the surface of Ribbon 1 was shiny and pristine with no sign of striation to the naked eye. This observation suggests a smoothing of the surface roughness due to material reflow during the stretching process. Exactly the same type of phenomenon is also observed during fiber drawing. No sign of crystallization or contamination was observed at the interface between TZN1 glass substrate and the TZN3 glass films. Another feature shown in Fig. 7(b) is the curvature of the ribbon surface with a minimum ribbon thickness in the central area. This is attributed to the non-homogeneous stretching ratio of Preform 1.

The thickness of the TZN3 glass optical guiding layer was measured to be 5.9 ± 0.5 μm in the central area. This latter value is lower than the 7 μm thickness expected from a stretch ratio of 1:5 for Preform 1 whose TZN1 glass layer minimum thickness was 34.9 ± 1 μm in the stretched region. Also no clear trend towards an increase of the TZN3 film thickness could be observed with respect to the ribbon length. The fluctuation and the non-homogeneities in stretching ratio mentioned above are thought to be the causes of the small discrepancy between experimental observations and theoretical predictions based on Preform 1 dimensions.

3.3.2. Ribbon 2

The features discussed above for Ribbon 1 were also observed for Ribbon 2. The thickness of the TZN2 and TZN3 optical guiding layers were measured to be 1.8 ± 0.2 μm and 5.6 ± 0.2 μm respectively. These values are each lower than 2 μm and 6 μm, respectively, which were expected from Preform 2 dimensions for a 1:5 ratio. The SEM analysis revealed neither signs of homogenous crystallization in the glass layer nor heterogeneous crystallization at the interface between the layers. This absence of crystallization is consistent with the relatively large Hruby stability parameter of the glasses (see Section 3.1) and the absence of crystallinity according to SEM imaging of the parent Preform 2 (see Section 3.2.2).

3.4. Optical assessment

Optical waveguiding at 632.8 nm wavelength was observed for both types of ribbon. This is illustrated in Fig. 8 for a 9 mm long waveguide sample prepared from Ribbon 2. It should be noted that, for this latter, the input beam coupled into both of the high refractive index layers TZN2 and TZN3. A more detailed investigation would be required to investigate coupling between the TZN3 and TZN2 glass layers or *vice versa*. This is out of the scope of the present report. As seen in Fig. 8, the exit face of the composite waveguide appeared rather uneven along the edges and this is certainly due almost entirely to the method of preparation of the waveguide endface (see Section 2.4), and there is potential for improving this.

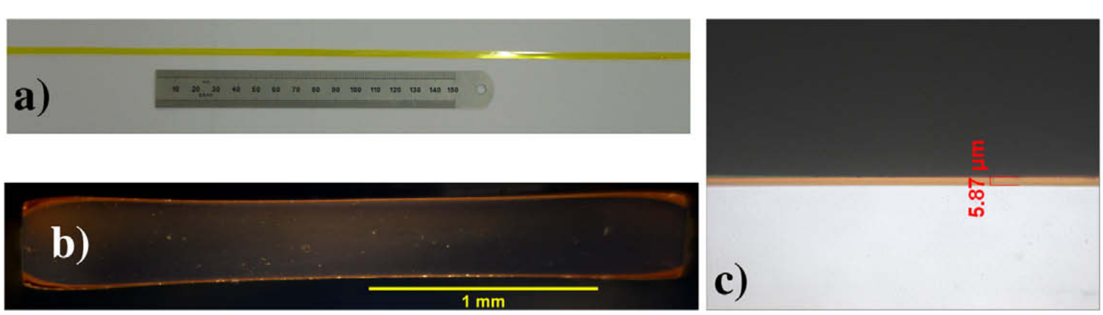

Fig. 7. (a) Plan view photography of Ribbon 1; (b) optical micrograph showing a cross-sectional view of Ribbon 1 (please note the presence of a double TZN3 glass film, thickening of the thickness towards the edges of the ribbon as well as the slight curved shape of the film surface) and (c) 40× magnification optical micrograph showing a cross-sectional view of the TZN3 glass optical guiding film on the TZN1 glass substrate in the central area of Ribbon 1.

4. Conclusions

The results presented demonstrate the feasibility of fabricating planar waveguides in ribbon format by stretching a preform made by extrusion. Such an approach has some major advantages over standard techniques, in its versatility of usable glass compositions, conservation of glass bulk stoichiometry and hence also of glass physical properties. Moreover, this route offers one-step manufacture of multi-layered waveguiding structures. The implementation of the process is simple, cost efficient, potentially enabling high productivity providing that lateral and longitudinal thickness variations of the layers in the extruded preform can be reduced. Further work on die design and extrusion conditions will be carried out towards these objectives.

In terms of format and dimensions, the width of the ribbon was limited to $\leqslant$3 mm because of the constraints on the preform width. However, such width dimension still allows integration of a number of parallel functionalities involving waveguiding structures of a few microns' width. Also, wider ribbon can readily be achieved by adjusting the preform and waveguide fabrication conditions. Nevertheless, it is not intended that the ribbon geometry would compete with a large dimensioned platform when fabricating highly complex optical circuitry. Instead, it should be seen as a low cost and versatile approach for the fabrication of a broad range of simple but high quality and compact optical devices in planar format. There is a real commercial prospect for soft glass planar waveguide in ribbon format from uniform preforms. Extrusion and other preform fabrication techniques will be further investigated to demonstrate this statement.

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